

03.06.24

End Term Examination for Even (Spring) Semester 2024
(For 1st Year 2nd Semester of 2-Year MCA)

Subject: Operating System
Full Marks: 80

Paper Code: MCAPGPC09
Duration: 3 hours

Group A
(Answer all)

10 x 1 = 10

1. Which memory is used to store PCB?
2. What are the disadvantages of multiprocessor system?
3. What is real time system?
4. Which process need to be synchronized.
5. What are the benefits of co-operating process?
6. What are the tasks of short-term scheduler?
7. When context switch is needed?
8. What is the bootstrap program?
9. Distinguish between CPU bound and I/O bound processes?
10. What is the meaning of term busy waiting?

Group B
(Answer any four)

4 x 5 = 20

1. Why CPU scheduling is needed? Distinguish between preemptive and non-preemptive scheduling. What the goal of CPU scheduling? 2+2+1
2. Explain race condition with suitable example. What are the different criteria that a solution to the critical section problem must satisfy? Name one classic problem of synchronization. 2+2+1
3. Explain process state transition diagram? Explain types of thread. 3+2
4. Explain mutual exclusion. Give the implementation of bounded buffer producer consumer problem using semaphore 2 + 3
5. What is paging? What is Page Table? Explain the conversion of Virtual Address to physical address in paging with suitable example and diagram. 1 + 4
6. Distinguish between contiguous and linked allocation of disk space. 1 + 2 + 2
7. List the advantage and disadvantage of two – level directory? Discover the ways to overcome the disadvantages of two – level directory. 1 + 1 + 3

Group C
(Answer any five)

5 x 10 = 50

1.
 - a. List the differences between CPU – bound and I/O – bound process. 2
 - b. Context switch is overhead to the system. Justify. 2
 - c. Consider the 6 processes whose arrival and burst time are as given in the table below.

Process Id	Arrival Time	Burst time	Priority
P1	1	4	2
P2	2	5	3
P3	2	7	3
P4	3	8(H)	5
P5	3	5	4
P6	4	6	2

Expectation
= 50/90

Calculate the average waiting time and turn-around time for priority scheduling for preemptive and non-preemptive case. 6

2.

- What are the uses of job queue, ready queue and device queue? Which memory stores all these queues? 3 + 1
- What we need from memory? Draw the memory hierarchy diagram showing all metric variation in it. 1 + 3
- Calculate the number of pages and internal fragmentation if page size is 4096 bytes and process size is 72,766 bytes. 2

3.

- List the duties of OS in connection with memory management. Briefly describe the address translation in contiguous allocation. 2 + 3
- Evaluate the maximum number of pages needed if a system support 32-bit address line and 512 MB page size. How the problem of internal fragmentation can be solved? 3 + 2

4.

- Can we apply optimal page replacement algorithm always? Justify your answer. 1 + 3
- What are frames? Given page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1. Compare the page faults for FIFO, LRU and Optimal Page replacement algorithm (Assume #frame = 3). 1 + 5

5.

- Briefly describe the different techniques to handle deadlock. 4
- Using Banker's algorithm answer the following for table given below.
 - How many resources of A, B, and C are there?
 - What are the contents of need matrix?
 - Find the safe sequence if it is in safe state. 6



Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	1	0	1	2	1	1	2	1	1
P1	2	1	2	5	4	4			
P2	3	0	0	3	1	1			
P3	1	0	1	1	1	1			

6.

- List the advantages and disadvantages of layered approach of Operating System? Explain the effect of increasing the time quantum to an arbitrary large number and decreasing the time quantum to an arbitrary small number for round robin scheduling algorithm with suitable example? 2 + 4
- Explain multiprogramming with fixed partition. Explain the difference between logical and physical addresses. 2 + 2

7.

- Explain first fit, best fit, worst fit and next fit with their advantages and disadvantages. 6
- What is thrashing? How it is controlled by OS. 4

E = 50/90

62 8 = 2 + 78

8 = 900000

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End Semester Examination (Spring Semester) 2024
(For 1st Year 2nd Semester MCA)

Paper Name: Computer Networks
Paper Code: MCAPGPC10

Full Marks: 80
Time: 3 hours

Group A (Answer all the questions)

5X2=10

1. Define multiplexing and demultiplexing.
2. What are the types of classful addressing?
3. Define HTTP.
4. Why are protocols needed in data communication?
5. Define throughput and jitter.

Group B (Answer any 5 questions)

5X6=30

1. What is transmission media? Explain different forms of transmission media. [1+5]
2. Describe the Pulse Code Modulation (PCM) technique with neat diagram. [6]
3. Differentiate:
i) ASK vs FSK ii) Synchronous TDM vs Statistical TDM iii) ARP vs RARP [2X3]
4. What is transmission impairment? Explain the various types of transmission impairments. [2+4]
5. What is the difference between network layer delivery and transport layer delivery? List some uses for UDP protocol. [3+3]
6. Describe the Message Transfer Agent of E-mail Communications. [6]

Group C (Answer any 4 questions)

4X10=40

1. What is topology? Describe any two physical topologies with their advantages and disadvantages. With neat diagrams explain various line coding techniques (Digital to digital conversion). [1+4+5]
2. What are the services provided by data link layer? Briefly explain the CSMA/CD method with neat diagram. Compare and contrast the Go-Back-N ARQ Protocol with Selective-Repeat ARQ. [2+5+3]
3. Briefly describe IP datagram header format with a neat diagram. In a block of addresses, we know the IP address of one host is 182.44.82.16/26. What is the first address (network address) and the last address (limited broadcast address) in this block? Briefly explain the protocols used in network layer. [4+2+4]

4. What are the services offered by TCP in the transport layer? Illustrate the Scenarios for establishing a connection using a Three-Way Handshake. [3+7]

5. Write short notes on (any four) : [2.5X 4]

- i) DNS
- ii) FTP
- iii) TELNET
- iv) MIME
- v) IMAP4

02.06.24

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Spring Semester Examination - 2024

MCA 1st year, 2nd semester Examination

Paper Name: ELECTIVE-I (Image Processing)

Paper Code: MCAPGPE02

Full Marks: 80

Time: 3 hrs

Group-A

Answer all questions

Marks: 10 x 1=10

1. What is an image gradient?
2. Differentiate image correlation and convolutional.
3. How many bits are required to store a 8x8 Image with maximum gray-level 64?
4. Explain the term LBP.
5. Analyze the names of color models in DIP.
6. Write expression for power-law Image transformation.
7. What is the homogeneity feature?
8. Write the kernel values for LoG.
9. Discuss Image smoothing technique.
10. How does the quantization technique work?

Group-B

Answer any Five (05) questions

Marks: 5 x 6=30

1. Define GLCM technique mathematically and derive the general formulation for deriving feature vectors using this method. 3+3
2. Explain each step of frequency domain Image filtering technique. 6
3. Write a short note on (i) Erosion, (ii) Dilation, (iii) Opening, and (iv) Closing Morphological Image Processing operator. 6
4. Explain the working principle of LBP with proper examples. Discuss why this scheme is used? 4+2
5. What is Image segmentation? Describe the various types of Image Thresholding technique. 2+4
6. Derive the formulation for 1st order and 2nd order Image Gradient filtering techniques. 3+3

Group-C

Answer any four (04) questions

Marks: 4 x 10=40

17. Define different color models in Digital Image Processing (DIP). Compute HSI components from the given R, G, and B components in color image processing with writing mathematical expressions for each H, S, I component. 4+6=10

R			G			B		
2	1	2	2	3	4	2	2	1
1	3	1	2	3	1	0	1	5
3	0	1	1	0	2	1	3	0

18. What are textures? What are the different methods of feature representation schemes from an image? Compute GLCM from the given image I for distance (d)=1, and angle=45 of pixel locations and compute Mxprob, contrast, correlation, and entropy. 2+3+5=10

2	1	3	2
0	0	2	0
3	2	2	1
0	2	1	2

19. Explain the design cycle of a pattern recognition system. Compute a Binary image from the given Gray-scale image by selecting a threshold 'T=7' using the global thresholding technique up to 2-iteration. 4+6=10

4	24	4	3
7	12	2	6
16	56	1	7
13	100	9	2

20. What are affine transformation techniques? Apply median filtering technique on the below image to find the unknowns in the output image. $3+7=10$

4	3	8	9
3	8	3	0
5	0	2	1
5	6	1	7

Input Image

4	3	8	9
3	X	Y	0
5	Z	W	1
5	6	1	7

Output Image

21. What are image compression techniques? Write short notes on "Discrete Wavelet Transformation", and "Discrete Cosine Transform" techniques mathematically. $2+4+4=10$

78/80
= 90

10.06.24

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Azhar Sir End-Semester Examination (Spring Semester) – 2024

(1st Year 2nd Semester 2-yrs MCA)

Subject Name: Data structures and Algorithms

Subject Code: MCAPGPC08

Full Marks: 80

Time: 3hrs

Group-A

(Answer all questions)

5 × 2 = 10

1. Why do we need data structure?
2. What is an Abstract Data type (ADT)? Explain it with example.
3. For the given infix notation $A+B*(C-D)/(P-R)$, find the prefix and post fix notation.
4. Define binary search tree with an example?
5. What is the output of bubble sort after the 2nd iteration given the following sequence?
24 56 47 35 10 90 82 31

Group-B

(Answer any six questions)

6 × 5 = 30

1. What is data structure? How data structures are classified? List the various operations that can be performed on data structure?
2. Write a C program to show the parameters passing in a C function.
3. Represent the following sparse matrix using link list.

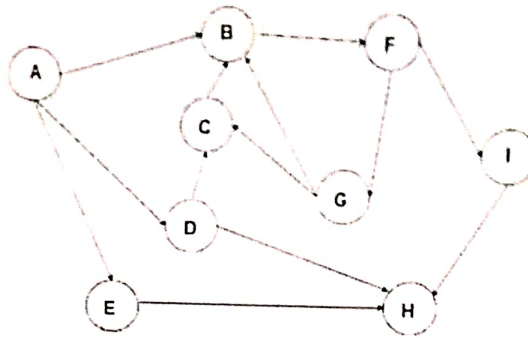
0	0	0	0	9	0
0	8	0	0	0	0
4	0	0	2	0	0
0	0	0	0	0	5
0	0	2	0	0	0

4. What are the limitations of queue? Explain with suitable example the insertion operation in a circular queue.

5. Sort the following elements [5, 3, 2, 6, 4, 1, 3, 7] step by step using quick sort.
6. Evaluate the following postfix expression $[548+*6/62/8*-]$ using stack.
7. The post-order and in-order traversals of a tree yield the following sequence of nodes:
 - Post-order: D G H E B F C A
 - In-order: D B G E H A C F

Draw the binary tree T and find out its Pre-order sequence.

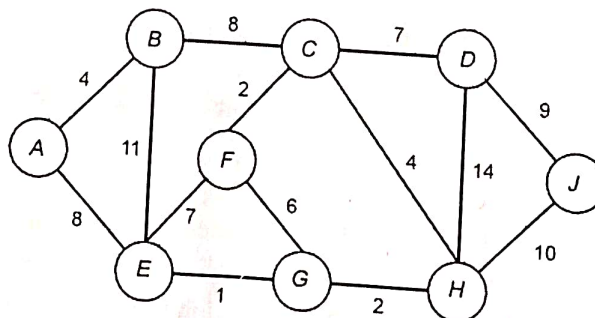
8. Traverse the following graph using breadth first search (BFS) and depth first search (DFS) algorithm.



Group-C
(Answer any four questions)

$4 \times 10 = 40$

1. What is linked list? Write and explain the algorithm for creation of a node and insertion of a node at the end of the singly list.
1 + 4 + 5
2. Perform the parenthesis checking of expression $[3 + \{5 + 8 * (4 - 3) * (6/3)\} - 2]$ using stack.
5 + 5
3. Convert the expression $(7 - 5) * (9/2)$ from Infix to Postfix using stack.
5 + 5
4. Consider the following sequence [24 56 47 35 10 90 82 31]
Write the contents of the array after 5 passes of the outermost loop of insertion sort.
1 + 2 + 7
5. Write the contents of the array after 5 passes of the outermost loop of selection sort.
5 + 5
6. What is queue? What are the operations of queue? Write an algorithm to insert an element into a queue.
1 + 9
7. Create a binary search tree for the following numbers start from an empty binary search tree [45, 26, 10, 60, 70, 30, 40]. Delete keys 10, 60 and 45 one after the other and show the trees at each stage.
5 + 5
8. What is balance factor in AVL tree? Construct an AVL tree using following numbers [12, 18, 17, 8, 11, 5, 4] into an empty tree.
5 + 5
9. Find out the minimum spanning tree of the following graph step by step using Prim's and kruskal's algorithms.



End – Semester Examination (Spring Semester) – 2023-24

[MCA – 1st Year, 2nd Semester]

Paper Name: Object Oriented Programming with Java

Full Marks: 80

Paper Code: MCAPGPC11

Time: 3 Hrs

GROUP A

[Answer any five questions]

[5 x 2 = 10]

1. i) What are the conventions for naming methods in Java?
- ii) Explain the importance of the Java Virtual Machine.
- iii) What do you mean by up-casting and down-casting? Give examples.
- iv) What are the advantages of organizing classes into packages?
- v) Explain the use of join () method in the context of multithreading in Java.
- vi) How Errors differ from Exceptions?

GROUP – B

[Answer any four questions]

[4 x 5 = 20]

2. i) How Java achieved architecture neutrality (platform independence) property?
- ii) How abstract class is different from an interface? [2+3]
- iii) Briefly state the fundamental properties of object-oriented programming language.
4. Create an interface Area with the following details:
Data Member: a) float PI = 3.14159F
Methods: a) float getArea (float a, float b)
Implement the above interface in two different classes Rectangle and Cone.
5. Write a Java program which can simultaneously run two threads with appropriate priorities after creating the threads by implementing a Runnable interface.
6. Explain the differences among local variables, instance variables, and class (static) variables with examples.

GROUP – C

[Answer any five questions]

[5 x 10 = 50]

7. i) What are the different access specifiers available in Java? State their access scope.
b) Write a Java Program to calculate the minimum among numbers for different combination of inputs with an overloaded min () method. [(1+4)+5]
8. Explain the 'static', 'final' and 'super' keywords with all their usages! Support explanation with a suitable Java program. [5+5]
9. i) What is the purpose of 'this' keyword in Java?
- ii) What is a constructor and what are the advantages of using it?

- c) Write a program to find the Area and Volume of a Cylinder by extending a Shape Class which contain the area () method [Note: the volume () method need not to be defined in the Shape class]. [4+1+5]

10. a) What is the purpose of the finally block in Java?

b) What are the different types of inheritances?

c) What do you mean by garbage collection?

d) What is the base class of all types of Exception? [2+5+2+1]

11. a) Explain the significances of the keywords in the statement public static void main (String args[]).

b) How static polymorphism differs from the dynamic polymorphism?

c) With an example differentiate pre- and post- decrement operator. [4+3+3]

12. a) Explain the different types of bit-wise operators in Java with proper example.

b) What is 'InterruptedException'?

c) Draw and explain the life cycle (state transition diagram) of a Thread. [4+2+4]

70

if i'll not get 70 P.Sir will give the part , else i'll .

64/80

14.06.24

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End Semester Examination - 2024

For 2nd Semester MCA Programme

Paper Name: Artificial Intelligence

Paper Code: MCPGPC12

Full Marks: 80

Time: 3 Hrs

Group - A

(10X1=10)

Answer all of the following

- i. What is the range of the logistic regression output?
a) $[-1, 1]$ b) $[0, 1]$ c) $(0, \infty)$ d) $(-\infty, \infty)$
- ii. Which method is used to estimate the parameters in logistic regression?
a) Ordinary Least Squares (OLS) b) Maximum Likelihood Estimation (MLE)
c) Gradient Descent d) Principal Component Analysis (PCA)
- iii. In SVM, what is the objective of maximizing the margin?
a) Increase model complexity b) Improve generalization to unseen data
c) Minimize training error d) Reduce the number of support vectors
- iv. Which kernel function is commonly used in SVM for non-linear classification?
a) Linear kernel b) Polynomial kernel
c) Radial Basis Function (RBF) kernel d) Sigmoid kernel
- v. Which metric is more concerned with avoiding false negatives?
a) Precision b) Recall c) F1-score d) Accuracy
- vi. What does the diagonal of a confusion matrix represent?
a) True positives and true negatives b) False positives and false negatives
c) True positives and false positives d) True negatives and false negatives
- vii. To find the minimum or maximum of a function using gradient descent, what step is taken?
a) Move in the direction of the gradient of the function.
b) Move in the direction opposite to the gradient of the function.
c) Set the gradient of the function to zero.
d) Set the second derivative of the function to zero.
- viii. What type of algorithm is A*?
a) Depth-first search b) Breadth-first search
c) Informed search d) Uninformed search
- ix. Which component of the PEAS framework describes the criteria for the success of an intelligent agent?
a) Performance b) Environment c) Actuators d) Sensors
- x. In fuzzy logic, the membership function assigns values between
a) -1 and 1 b) 0 and 1 c) 0 and infinity d) 1 and infinity

Group - B
(Answer any 4 questions)

(4X5=20)

1. What is confusion matrix?
2. Difference between supervised and unsupervised learning- explain.
3. Define precision and recall.
4. What is over-fitting and under-fitting?
5. What is the importance of AUC- ROC in Machine Learning?
6. Difference between Weak AI and Strong AI.
7. Difference between BFS and DFS.

Group - C
(Answer any five questions)

(5X10=50)

8. How are Artificial Neural Networks different from Normal Computers? Explain Heuristic Search Algorithm. Name various type of agents. 4+4+2
9. Write down the working principle of A* search algorithm. Discuss the characteristics of fuzzy logic. Difference between crisp set and fuzzy set- explain. 4+3+3
10. How linear regression is used to model the relationship between a dependent variable and one or more independent variables. Describe the difference between simple linear regression and multiple linear regression. 5 + 5
11. Explain the basic concept of decision tree. What are the advantages and disadvantages of decision tree. 5+ 5
12. How does SVM work for classification and regression tasks? Write down some names of kernel functions in SVM. What are the advantages of SVM. 5+2+3
13. Write down the algorithm K- Nearest Neighbors. What are the disadvantages of KNN algorithm. How to choose the value of k for KNN Algorithm? 5+2+3
14. Briefly describe the working principle of K-means clustering algorithm. Give an example of k-means in real life. Write down the difference between Classification and Clustering. 5+2+3
15. Short Note (any two):
 - A) BFS
 - B) Fuzzy set
 - C) Actuator and Sensors
 - D) DFS 5*2